

ENSC 225 - Lab #2

Characterization of Diodes and Simple Diode Circuits

Ryan Thomas - 

Nadevni Waduge - 

Dawson Wakelam - 

Abstract

This lab report outlines the findings from ENSC 225 Laboratory #2, which investigates the characterization of diodes and the functionality of simple diode circuits. The experiment encompasses a comprehensive analysis of both signal and Zener diodes, focusing on their I-V characteristics, scale current, and technology factor. Participants collected and plotted forward and reverse characteristics, determining small signal resistances at various forward voltages through carefully constructed circuits. These measurements enabled the calculation of Zener voltage and resistance, enhancing understanding of diode behavior in practical applications.

In addition to the characterization of diodes, the lab included the construction and analysis of various rectifier circuits, including a simple half-wave rectifier, precision half-wave rectifier, and bridge rectifier. Through these experiments, students observed and recorded input and output waveforms, measured ripple voltage, and compared the performance of the different rectification methods. The report emphasizes the importance of accurate measurements and theoretical comparisons, presenting data that reflects both experimental outcomes and manufacturer specifications. Overall, this lab provided a hands-on approach to understanding diode operation and rectifier design in electronic circuits.

Experiment 1

Introduction: This section of the lab focuses on analyzing the current-voltage (I-V) characteristics of P-N diodes, specifically signal and Zener diodes. Understanding these characteristics is essential for their applications in electronics, such as rectification and voltage regulation. Using the Specialized Measurement Apparatus (SPA), we will collect data on the forward characteristics of both types of diodes and the reverse characteristics of the Zener diode. The collected data will be plotted to visualize their operational behavior, highlighting the relationship between voltage and current under different biasing.

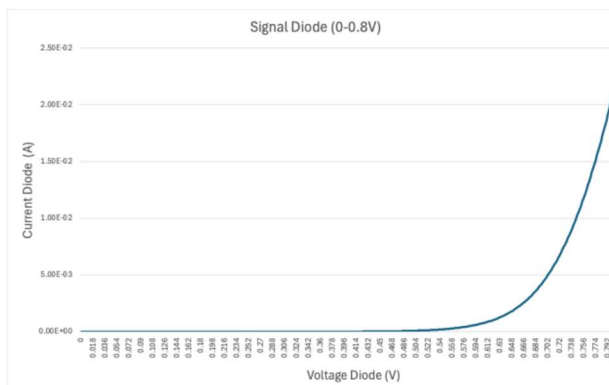


Figure 1- Forward Signal Diode:

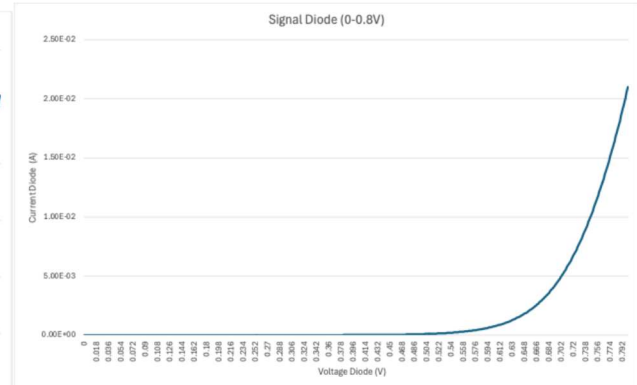


Figure 2 - Forward Zienr Diode

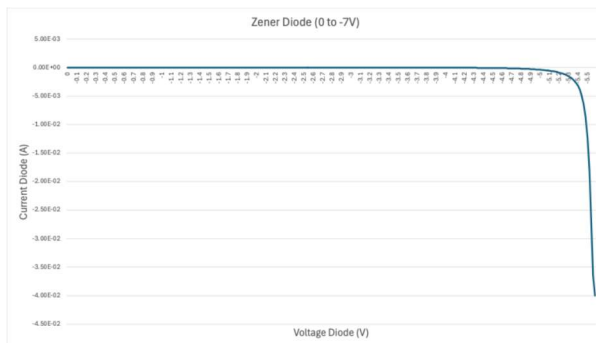


Figure 3 - Backwards Zienr Diode

Discussion:

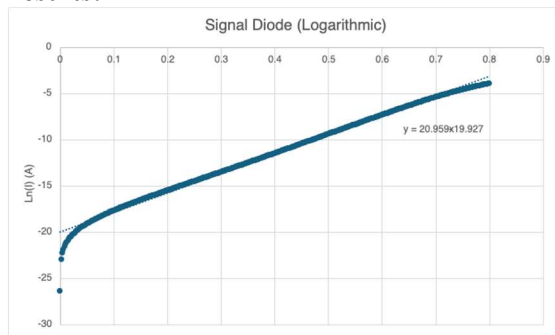
The results demonstrate typical I-V characteristics for both the signal and Zener diodes. The signal diode shows an exponential increase in forward current starting at approximately 0.6V, with significant current (0.020983 A) at 0.8V, indicating efficient conduction. In contrast, the Zener diode displays a lower forward current, reaching only 0.012769 A at the same voltage. The presence of small negative currents at lower voltages suggests reverse leakage, common in Zener diodes.

In reverse bias, the Zener diode exhibits a clear breakdown behavior, with substantial negative current as the voltage approaches the breakdown region (around -5V to -7V). This characteristic allows the Zener diode to maintain a stable voltage while conducting reverse current, highlighting its ability to regulate voltage effectively. Overall, the data aligns with theoretical expectations, showcasing the distinct functionalities of each diode in both forward and reverse bias conditions.

Experiment 2:

Introduction: This section will determine the scale current and technology factor (n) for a signal diode. The experiment involves plotting $\ln(I_D)$ against V_D and fitting a line to the linear portion (100 pA to 1.5 mA). This analysis will extract the scale current and n, with additional guidance from the "Ln Plot" document.

Results:



Scale Current	Technology Factor (n)
$\ln(I_s) = -19.927$ $(I_s) = e^{-19.927}$ $(I_s) = 2.22 \times 10^{-9} \text{ A}$	$20.959 = \frac{1}{\eta V_T}$ $20.95 = \frac{1}{\eta(25.69 \times 10^{-3})}$ $\eta = 0.544$

Discussion:

The graph in the Results section shows the linearized portion of the signal diode's characteristics with the natural logarithm of the current. From the fitted straight line, the scale current and technology factor (n) are determined to be $2.22 \times 10^{-9} \text{ A}$ and 0.544, respectively.

Experiment 3:

Introduction: In this section, the small signal resistances of a signal diode were determined at various forward voltages. The team selected three pairs of resistors (1k, 12k, and 100k) and measured their resistance values to construct a circuit as shown in Schematic-1. After applying a 10V DC supply, they calculated the diode voltage and the DC current (I_{D1}). The diode was then shunted with another 1k resistor to obtain a new voltage, allowing for the determination of small signal resistance at I_{D1} . This process was repeated for the 12k and 100k resistors to find I_{D2} and I_{D3} . Finally, the team compared the small signal resistances obtained experimentally, graphically, and through calculations.

Results:
Measured Resistors:

	1 k Ω	12 k Ω	100 k Ω
Pair 1	0.983	11.62	98.50
Pair 2	0.987	11.78	98.17

R = 1 k Ω	V_{DI}	I_{DI}
Without Shunt	0.7273 V	$\frac{V_s - V_D}{R} = \frac{10.00 - 0.7273}{0.983} = 9.4331 \text{ mA}$
With Shunt	0.7194 V	$\frac{V_s - V_{Shunt}}{R} - \frac{V_{Shunt}}{R_{Shunt}} = \frac{10.00 - 0.7194}{0.983} - \frac{0.7194}{0.987} = 8.7122 \text{ mA}$

R = 12 k Ω	V_{DI}	I_{DI}
Without Shunt	0.6008 V	$\frac{V_s - V_D}{R} = \frac{10.00 - 0.6008}{11.62} = 0.8103 \text{ mA}$
With Shunt	0.5970 V	$\frac{V_s - V_{Shunt}}{R} - \frac{V_{Shunt}}{R_{Shunt}} = \frac{10.00 - 0.5970}{11.62} - \frac{0.5970}{11.78} = 0.7585 \text{ mA}$

R = 100 k Ω	V_{DI}	I_{DI}
Without Shunt	0.4977 V	$\frac{V_s - V_D}{R} = \frac{10.00 - 0.4977}{98.50} = 0.09647 \text{ mA}$
With Shunt	0.4941 V	$\frac{V_s - V_{Shunt}}{R} - \frac{V_{Shunt}}{R_{Shunt}} = \frac{10.00 - 0.4941}{98.50} - \frac{0.4941}{98.17} = 0.09147 \text{ mA}$

Graphical calculations for small signal resistance:

$$\frac{I_{shunted\ 1} - I_{shunted\ 2} - I_{shunted\ 2}}{V_{shunt\ 1} - V_{shunted\ 2} - V_{shunted\ 2}} = \frac{8.7122 - 0.7585 - 0.09147}{0.7194 - 0.5970 - 0.4941} = 21.152$$

Small-Signal Resistance				
		Experimental	Graphical (Excel)	Using Formula ($r_z = \frac{\eta V_T}{I_D}$)
r _{d1}	I _{D1}	$\frac{0.7273 - 0.7194}{9.4331 \times 10^{-3} - 8.7122 \times 10^{-3}}$	$\frac{1}{21.152} \times \frac{0.7194}{8.7122 \times 10^{-3}}$	$\frac{0.544(25.69 \times 10^{-3})}{9.4331 \times 10^{-3}} = 1.4815\Omega$
	I _{D1(shunt)}	= 10.9585 Ω	= 3.9038 Ω	$\frac{0.544(25.69 \times 10^{-3})}{8.7122 \times 10^{-3}} = 1.6041\Omega$
r _{d2}	I _{D2}	$\frac{0.6008 - 0.5970}{0.8103 \times 10^{-3} - 0.7585 \times 10^{-3}}$ = 73.3591 Ω	$\frac{1}{21.152} \times \frac{0.5970}{0.7585 \times 10^{-3}}$	$\frac{0.544(25.69 \times 10^{-3})}{0.8103 \times 10^{-3}} = 17.2471\Omega$
	I _{D2(shunt)}		= 37.2107 Ω	$\frac{0.544(25.69 \times 10^{-3})}{0.7585 \times 10^{-3}} = 18.4250\Omega$

r _{d3}	I _{D3}	$\frac{0.4977 - 0.4941}{0.09647 \times 10^{-3} - 0.09147 \times 10^{-3}} = 1120\Omega$	$\frac{1}{21.152} \times \frac{0.4941}{0.09147 \times 10^{-3}} = 255.3787\Omega$	$\frac{0.544(25.69 \times 10^{-3})}{0.09647 \times 10^{-3}} = 144.8674\Omega$
	I _{D3(shunt)}			$\frac{0.544(25.69 \times 10^{-3})}{0.09147 \times 10^{-3}} = 152.7863\Omega$

Discussion:

In this section, the team successfully determined the small signal resistances of a signal diode at various forward voltages using different resistor pairs. The measured resistance values for the 1k, 12k, and 100k resistors were consistent with their nominal values, confirming the accuracy of the setup. After applying a 10V DC supply, the calculated diode currents (ID1, ID2, ID3) were effectively utilized to derive small signal resistances. The experimental small signal resistances varied significantly, with values of approximately 10.96Ω, 73.36Ω, and 1120Ω for ID1, ID2, and ID3, respectively. Graphical calculations and formula-based approaches yielded comparable results, highlighting a small resistance for ID1 and an increasing trend for higher currents. This behavior aligns with the expected characteristics of diodes, demonstrating the relationship between forward current and dynamic resistance. Overall, the results reinforce the principles of diode behavior in small signal applications and underscore the effectiveness of the measurement techniques employed.

Experiment 4

Introduction: In this part of the lab, we construct a similar circuit to experiment 3, here we look to find the output Zener voltage and current through measuring the (Zener) diode voltage by using a 1K Ohm resistor and then shunting the Zener diode with a 10K Ohm resistor and measuring the diode voltage again. With these measurements we can take advantage of the equation: $V_Z = V_{Z0} + I_Z r_Z$ in order to determine the Zener diodes output voltage and resistance.

Results:

Resistor	Exact Measurement
1kΩ	0.9867
10kΩ	9.867

1kΩ	I _{Z1}	V _{Z1}
Without shunt resistor	$\frac{V_S - V_Z}{R} = \frac{10.015 - 5.439}{0.9867} = 4.637681$	5.439

10kΩ	I _{Z1}	V _{Z1}
With 10KΩ shunt resistor	$\frac{V_S - V_{ZS}}{R} - \frac{V_{ZS}}{R_S} = \frac{10.015 - 5.428}{0.9867} - \frac{5.428}{9.867} = 4.09871$	5.428

Using the formula $V_Z = V_{Z0} + I_Z r_Z$ where the Z in V_Z and I_Z can be replaced by Z1 and Z2 respectively we can subtract the equations from each other and end up with:

$$V_{Z1} - V_{Z2} = (I_{Z1} - I_{Z2})r_Z$$

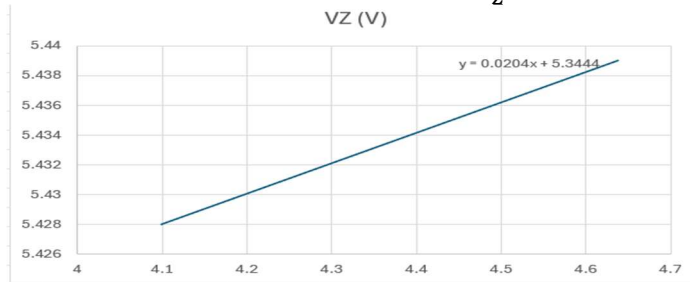
Plugging in values we get:

$$(5.439 - 5.428) = (4.637681 - 4.09871) r_Z$$

$$r_Z = 0.0204093k\Omega \text{ or } 20.41\Omega$$

Graphically through excel we get our slope equation to be $y = 0.0204x + 5.3444$

$$r_Z = 0.0204k\Omega = 20.4\Omega$$



IZ (mA)	VZ (V)
4.6377	5.439
4.0987	5.428

Discussion: The intention of this experiment was to observe and determine the characteristics of the zener diode and then compare the graphical and experimental approaches to ensure cohesiveness. Our results showed extremely close similarities between our graphical and experimental approaches at 20.4 and 20.41 ohms respectively suggesting an accurate and successful experiment, and the diode is functioning and behaving as expected.

Experiment 5

Introduction: The general idea in section 5 of the lab is to observe and analyze both input and output wave forms of a simple half wave rectifier circuit. We will add a capacitor to filter the output and give us a clean DC voltage value, and finally significantly lower the frequency in order to produce and observe a ripple wave and get a ripple voltage value.

Results:

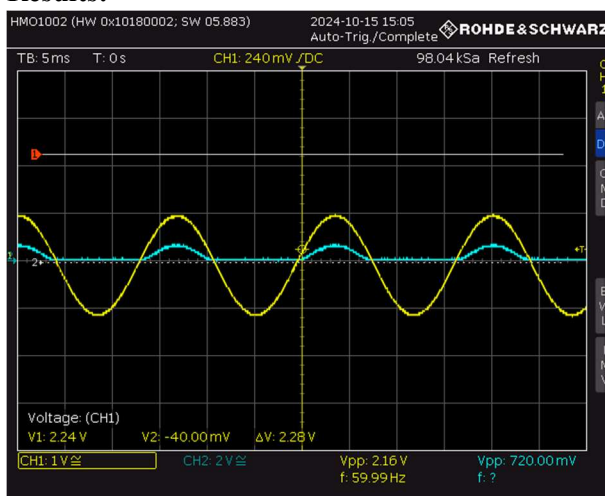


Figure 1 - Output of 1V Peak Sinusoidal Input

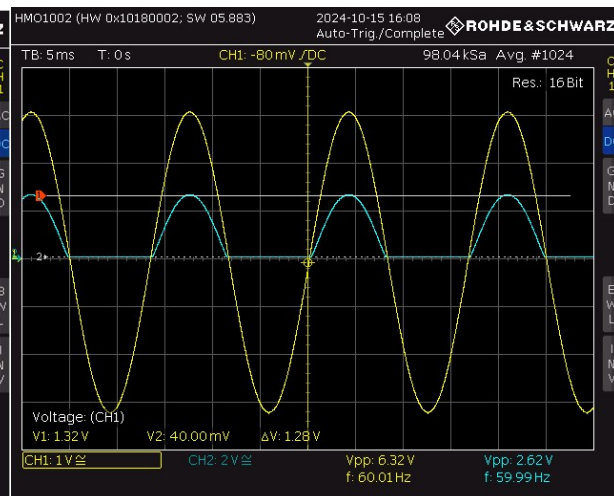


Figure 2 - Output of 3V Peak Sinusoidal Input

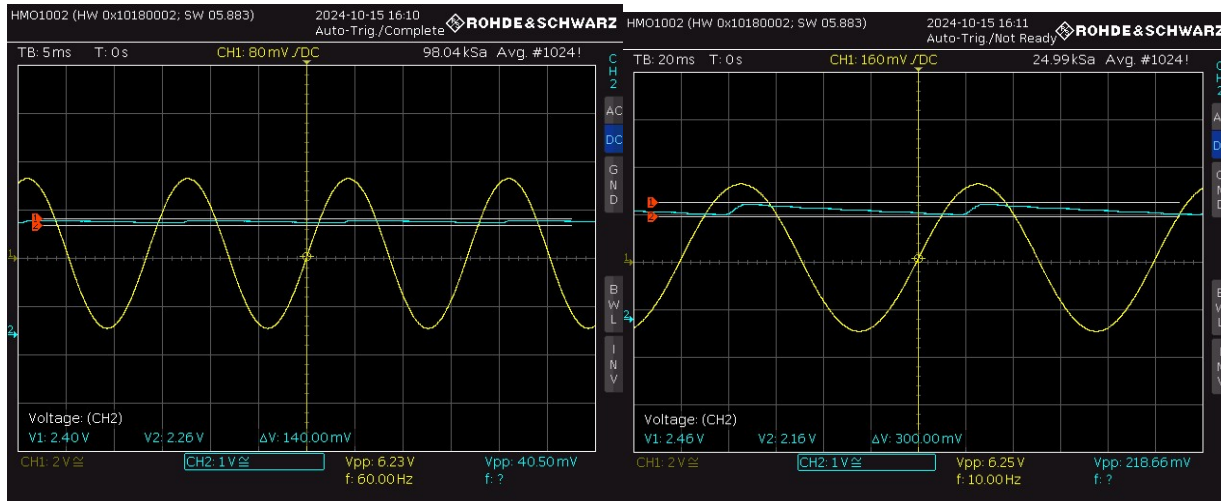


Figure 3 - Output of 100uF Electrolytic Capacitor

Figure 4 - Output of Capacitor @ 10Hz (Low) Frequency

Measured Resistors		
R1 10kΩ	R2 1kΩ	R3 10kΩ
9.867kΩ	0.9867kΩ	9.910 kΩ

3V Peak Cursor Measure Measurements		
Input Amplitude	Output Amplitude	Diode Voltage Drop
3.06V	2.56V	0.50V

$$V_{DC} = 2.40V \quad V_{RPP} = 0.300$$

$$Ripple\ Factor = \frac{0.300V}{2.40} = 0.125$$

Discussion:

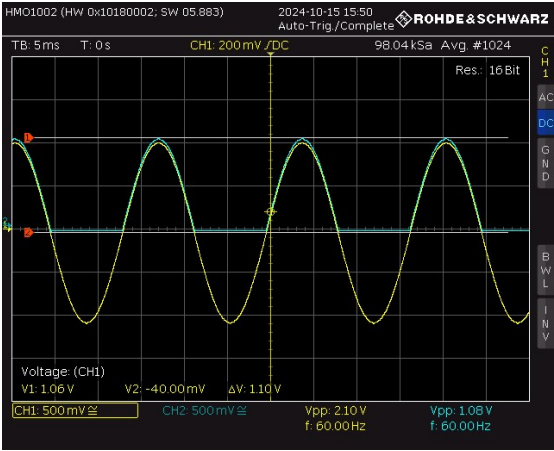
The purpose of a rectifier is to convert AC input signals to DC output signals. Observing this behaviour, we can see that when we set the function generator to a 3V input, the measured input amplitude will be 3.06V, the output amplitude will be 2.56V, resulting in a 0.50V voltage drop due to the conversion. Simple half wave rectifiers are unable to rectify AC signals with low amplitudes, this is known as the cut-off voltage limitation and is why they are mostly used in applications where the input voltage I significantly high.

Experiment 6

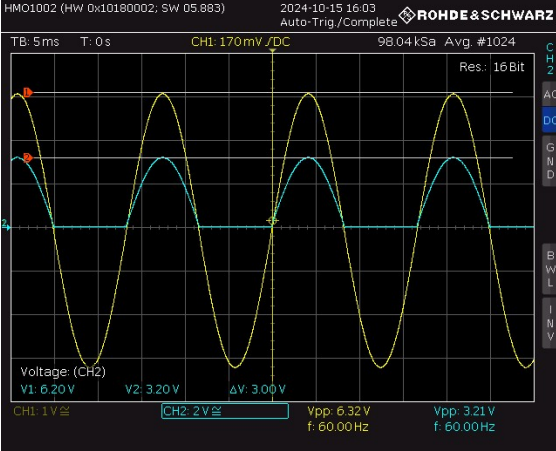
Introduction: In this section, we will explore a precision half-wave rectifier circuit that eliminates the diode's voltage drop, providing accurate rectification even at low signal levels. Using *Schematic 4*, the circuit is designed with an operational amplifier to improve performance compared to standard rectifiers. The experiment will focus on constructing the circuit, capturing output waveforms, and evaluating its performance, particularly the lowest peak-to-peak signal it can accurately rectify.

Results:

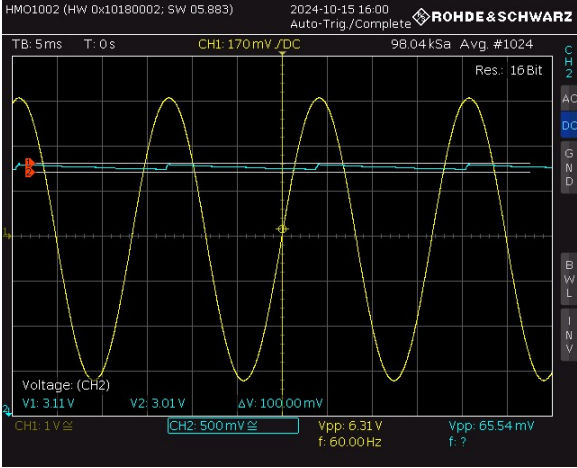
Output with 1V Peak Sinusoidal Input



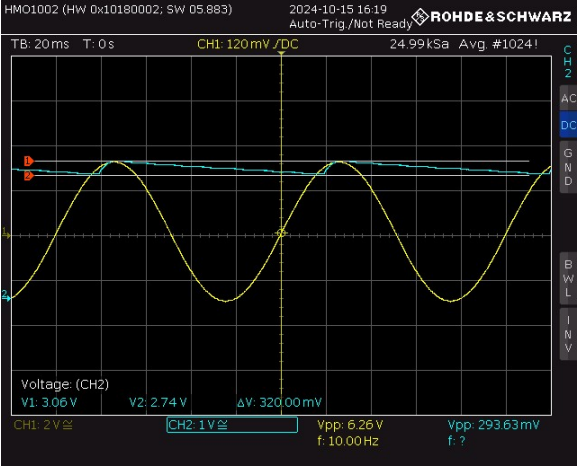
Output with 3V Peak Sinusoidal Input



Output with 100uF Electrolytic Capacitor



Output with Capacitor at 10Hz Freq



Measured Resistors		
R1 10kΩ	R2 1kΩ	R3 10kΩ
9.867kΩ	0.9867kΩ	9.910 kΩ

3V Peak Cursor Measure Measurements		
Input Amplitude	Output Amplitude	Diode Voltage Drop
3.14V	3.10V	0.04V

$V_{DC} = 3.11V$

$V_{RPP} = 0.320V$

$Ripple\ Factor = \frac{0.320V}{3.11V} = 0.103$

Discussion:

Comparison Between Simple and Precisions values

	Simple Half-wave Rectifier	Precision Half-wave Rectifier
Input Amplitude	3.06V	3.14V
Output Amplitude	2.56V	3.10V
Diode Voltage Drop	0.50V	0.04V
V_{DC}	2.40V	3.11V
V_{RPP}	0.300V	0.320V
Ripple Factor	0.125	0.103

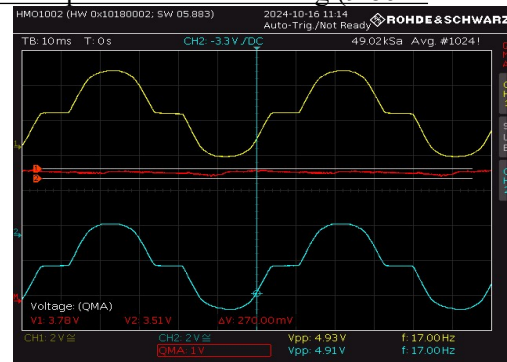
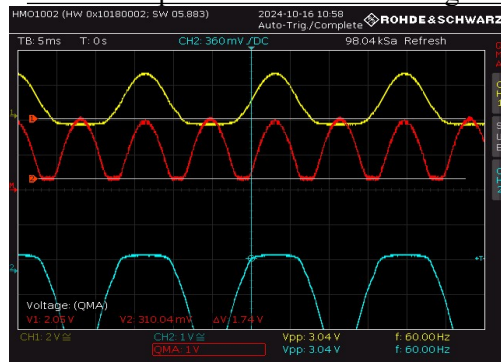
This experiment allows us to see the difference between a simple and precision half-wave rectifier. The precision half-wave rectifier has peak input and out voltage values of 3.14V and 3.10V, which are almost the same. This results in the diode voltage drop being 0.04V, which is almost zero. The circuit for the precision is designed to overcome the voltage drop across the diode. Adding the capacitor allows for a smoother output wave, giving us a DC voltage value of 3.11V. When the frequency is 10Hz, the ripple voltage is measured to be 0.320V, leading to a ripple factor of 0.103. The precision half wave rectifier allows us to rectify AC signals with very low amplitudes.

Experiment 7

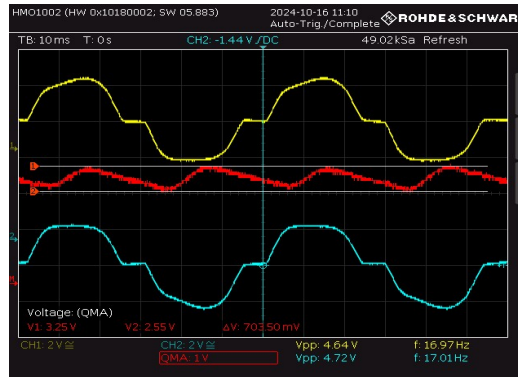
Introduction: In this section, we will experiment with a bridge rectifier circuit using four signal diodes. The goal is to observe and analyze the full wave rectified output waveform. Initially, the circuit will be tested without a filter capacitor to view the unfiltered rectified output. Afterward, a 100 μ F filter capacitor will be added to smooth the output into a more stable DC voltage. We will compare both the filtered and unfiltered waveforms and capture data to evaluate the bridge rectifier's performance.

Results:

Rectifier Operation without Filtering of 100uFD Rectifier Operation with Filtering (9.862k Ω Resistor)



Rectified Operation with Filtering (0.9842k Ω Resistor)



Unfiltered Bridge Rectifier	
Input Peak-Peak	3.04V
Output Amplitude	1.74V

Filtered Bridge Rectifier	
Input Peak-Peak	4.64V
Output (DC)	4.72V
Ripple Voltage Peak-Peak	0.704V

In the unfiltered circuit (left), the output isn't as smooth because there's no op-amp or other active components to stabilize it. However, when a $100\mu\text{F}$ capacitor is added in parallel with the load resistor ($0.9842\text{k}\Omega$), as shown in the right figure, the output improves significantly. The capacitor helps smooth the waveform, making use of both halves of the input signal and reducing the ripple. We tested using both a $10\text{k}\Omega$ load resistor and a $1\text{k}\Omega$ load resistor, and we determined that the 1k load provided us with a clearer waveform that is easier to measure. This is why we have 3 screen captures in total to show the difference between the two loads.

Using the values collected we can calculate the ripple factor, which is close to the other two.

$$r = \frac{0.704}{4.72 - 0.5(0.704)} = 0.161$$

Conclusion

This lab provided hands-on experience with diodes and rectifier circuits, key components in electronic systems. By characterizing signal and Zener diodes using an SPA, we gained insight into their I-V behavior, confirming parameters like scale current and Zener voltage, especially in reverse breakdown and voltage regulation. The rectifier experiments, including simple and precision half-wave rectifiers and bridge rectifiers, demonstrated how AC signals are converted to DC. The precision rectifier, with its negligible voltage drop, was effective for low-signal rectification, while the bridge rectifier delivered efficient full-wave rectification. Adding a capacitor reduced ripple voltage, showing the importance of filtering for smooth DC output. Overall, the lab reinforced theoretical concepts and provided practical insights into diode behavior and rectifier design, while also giving valuable experience with the SPA.